SIMATS ENGINEERING
ASSESSMENT TOOL 2

COURSE CODE: ITA0605 **COURSE NAME: MACHINE LEARNING**

COURSE OUTCOME COVERED

CO3: Bayes Theorem – Concept Learning – Maximum Likelihood – Minimum Description Length Principle – Bayes Optimal Classifier – Gibbs Algorithm – Naïve Bayes Classifier – Bayesian Belief Network – EM Algorithm – Probability Learning – Sample Complexity – Finite and Infinite Hypothesis Spaces – Mistake Bound Model, (BL 6)

Assessment Tool Weightage: 20%

QUESTIONS: 15 **Total Marks:25**

Annexure A

Case Study

Code of Conduct:

I Cheredla Jasvina/192324026 certify that this submission is my original work and that I have adhered to the guidelines specified for this assessment.
I understand that any violation of academic integrity rules will result in disciplinary action.

Signature of the student:

Annexure A

Short Answer Test

2. **An organization wants to build a spam detection system for emails using machine learning techniques. Using Naïve Bayes and Bayes Optimal Classifier concepts, design a classification system. Compare Maximum Likelihood estimation with Minimum Description Length (MDL) principle in feature selection. Analyze system performance, discuss trade-offs in model complexity, and evaluate how sample size and hypothesis space affect classification accuracy.**
   Answer :

Spam Detection System Using Naïve Bayes and the Bayes Optimal Classifier

2.1 Problem Framing

An email spam filter is a binary classifier that assigns each incoming message to one of two classes, spam or ham (legitimate), based on observable features such as word occurrences, sender reputation, presence of links or attachments, and header metadata. This is a natural fit for probabilistic learning because the true label is uncertain given the evidence, and the system must reason under that uncertainty rather than apply rigid rules.

2.2 System Design Using Naïve Bayes

Feature representation: Each email is converted into a feature vector, typically using a bag-of-words or bag-of-tokens model, where each feature represents the presence or frequency of a particular word, phrase, or metadata attribute (e.g., "free", "winner", "click here", excessive capitalisation, number of hyperlinks).

Naïve independence assumption: Naïve Bayes assumes that, conditioned on the class label C (spam or ham), the features are mutually independent. This lets the joint likelihood be written as a product of per-feature likelihoods rather than requiring an intractable joint distribution over all word combinations.

Formally, for class c and feature vector (f1, f2, …, fn):

$$P(c \mid f1,\dots,fn) \propto P(c) \cdot \Pi\, P(fi \mid c)$$

Training: For each class, the system estimates P(c) as the relative frequency of spam versus ham in the training corpus, and P(fi | c) as the frequency with which feature fi appears in emails of class c (with Laplace/add-one smoothing to avoid zero probabilities for unseen words).

Classification: A new email is classified by computing the posterior score for both classes and choosing the class with the higher score — the Maximum A Posteriori (MAP) decision rule.

2.3 The Bayes Optimal Classifier

While Naïve Bayes commits to a single hypothesis (one estimated probability model), the Bayes Optimal Classifier (BOC) is the theoretical gold standard: instead of picking the single most probable hypothesis, it combines the predictions of every hypothesis in the hypothesis space, weighted by how probable each hypothesis is given the training data:

$P(c \mid E) = \Sigma\_h\ P(c \mid h) \cdot P(h \mid E)$, *summed over all hypotheses h in H*

No other classifier using the same hypothesis space and prior information can outperform the Bayes Optimal Classifier on average — it is provably optimal. In practice, summing over every possible hypothesis (every possible spam/ham probability model) is computationally infeasible, so Naïve Bayes is used as a tractable approximation: it effectively picks the single dominant hypothesis under a strong independence assumption instead of integrating over the full hypothesis space. Ensemble methods (e.g., combining Naïve Bayes with logistic regression and an SVM, and voting/weighting them by validation accuracy) approximate the spirit of the Bayes Optimal Classifier more closely than any single model.

2.4 Maximum Likelihood Estimation vs Minimum Description Length in Feature Selection

Aspect	Maximum Likelihood Estimation (MLE)	Minimum Description Length (MDL)
Objective	Choose parameters/features that maximise P(data | hypothesis)	Choose the hypothesis that minimises total description length: L(h) + L(D | h)
View of complexity	Ignores model complexity — more features almost always increase likelihood on training data	Explicitly penalises complexity — a hypothesis that needs many bits to describe is charged for it
Overfitting risk	High; tends to include noisy or spurious word features (e.g., rare tokens that happen to co-occur with spam by chance)	Low; naturally performs feature selection by discarding features that do not reduce the total encoding cost
Interpretation	"Best fit to the observed data"	"Best trade-off between model simplicity (Occam's razor) and fit to the data"
Spam filter effect	Larger, denser feature vocabulary; better training accuracy but weaker generalisation to novel spam wording	Smaller, more robust vocabulary of highly informative tokens; better generalisation to unseen spam campaigns

In practice, MLE is used to estimate the per-feature probabilities P(fi | c) once a feature set is fixed, while MDL (or MDL-inspired criteria such as mutual information thresholds, chi-squared pruning, or L1-regularised feature selection) is used upstream to decide which features are worth including at all. MDL formalises Occam's razor: among competing spam-detection models, prefer the one that compresses the training data most economically, since an overly complex hypothesis that memorises training quirks needs a long description and pays for it in the L(h) term.

2.5 Performance Analysis and Model-Complexity Trade-offs

- **Precision vs recall trade-off:** a spam filter that is too aggressive raises false-positive risk (legitimate mail marked spam), which is usually far costlier to users than a missed spam email (false negative). Threshold tuning on the posterior probability lets the system be biased toward high precision.
- **Model complexity:** a richer feature space (bigrams, header analysis, sender graphs) captures more signal but increases variance and training cost, and can violate the independence assumption more severely, degrading the calibration of Naïve Bayes probabilities even when the final class decision remains accurate.
- **Regularisation / smoothing:** Laplace smoothing acts as an implicit complexity control, preventing the model from being over-confident about rare words seen only a handful of times.
- **Concept drift:** spammers adapt their vocabulary, so a model tuned only for MLE fit to historical data degrades over time; periodic retraining or online Bayesian updating is needed to keep the estimated probabilities current.

2.6 Effect of Sample Size and Hypothesis Space on Accuracy

Sample size: With small training sets, probability estimates for P(fi | c) are noisy and dominated by sampling variance, so the MAP hypothesis chosen by Naïve Bayes can diverge substantially from the true Bayes-optimal decision. As the training set grows, the estimated posterior converges toward the true distribution (by the law of large numbers), and Naïve Bayes's decision approaches that of the Bayes Optimal Classifier, since with enough data the single MAP hypothesis increasingly dominates the hypothesis-averaged posterior.

Hypothesis space size: A larger hypothesis space (e.g., allowing arbitrary feature interactions instead of assuming independence) can in principle represent the true spam/ham distribution more accurately, but it requires proportionally more data to estimate reliably — this is the classic bias–variance trade-off. Naïve Bayes deliberately restricts the hypothesis space (via the independence assumption) to trade some representational bias for a large reduction in variance, which is why it performs surprisingly well even with modest training data, despite its assumption being technically false for natural language.

In summary, the spam filter should be understood as a practical, data-efficient approximation (Naïve Bayes) to an intractable theoretical ideal (the Bayes Optimal Classifier), with MDL-guided feature selection controlling model complexity, MLE handling parameter estimation within the chosen feature set, and sample size governing how closely the practical system can approach optimal performance.

3. **A telecommunications company aims to predict customer churn using historical usage data and service feedback. Using probabilistic models such as Naïve Bayes, Bayesian inference, and hypothesis space search, design a predictive system. Analyze how prior probabilities and feature dependencies influence churn prediction. Evaluate how uncertainty in customer behavior is handled and discuss the effectiveness of the model in improving customer retention strategies.**
   Answer :

Customer Churn Prediction Using Probabilistic Models

3.1 Problem Framing

Customer churn prediction is the task of estimating, for each customer, the probability that they will discontinue service within a given horizon, based on historical usage data (call minutes, data consumption, billing history, complaint frequency, contract type) and service feedback (satisfaction scores, support-ticket sentiment). The output is used to trigger proactive retention interventions before the customer actually leaves.

3.2 System Design

Feature set: Usage-pattern features (declining call/data usage, increased dropped calls), account features (tenure, contract type, billing disputes, price plan changes), and feedback features (support ticket sentiment, satisfaction survey scores, complaint counts).

Naïve Bayes churn model: Treats churn (C = yes/no) as the class variable and computes P(C | usage, billing, feedback) ∝ P(C) · Π P(featurei | C), estimating each conditional probability from historical churn records. This gives a fast, interpretable baseline that ranks customers by churn risk.

Bayesian inference layer: Rather than a single point estimate, the system maintains a probability distribution over churn risk that is updated incrementally as new usage data and feedback arrive, using Bayes' rule to combine the prior belief with new evidence: P(C | new evidence) ∝ P(new evidence | C) · P(C | prior evidence). This online-updating view is well suited to churn, since a customer's risk genuinely changes month to month.

Hypothesis space search: Beyond the naïve conditional-independence hypothesis, the system searches a broader hypothesis space of models — e.g., Bayesian networks with a small number of explicit dependency edges, or logistic-regression / tree-based hypotheses — evaluated by cross-validated posterior predictive performance, effectively searching for the hypothesis (or hypothesis class) that best balances fit and generalisation, analogous to a Bayesian model-selection procedure over candidate structures.

3.3 Role of Prior Probabilities

The prior P(C) — the base churn rate in the customer population — matters a great deal in this domain because churn is typically a rare-event problem (e.g., 2–5% monthly churn). A model that ignores the prior and only looks at likelihood ratios can badly misjudge risk for an average customer; Bayesian scoring correctly down-weights weak evidence when the prior probability of churn is already low, which limits false alarms. Priors can also be set per segment (e.g., higher base churn prior for month-to-month contracts than for two-year contracts), letting the model start each customer with a more informed baseline before individual evidence is incorporated.

3.4 Feature Dependencies

Real churn drivers are rarely independent: a billing dispute often triggers a support ticket, which drives down satisfaction score, which in turn correlates with reduced usage — the naïve independence assumption undercounts this compounding effect. Modelling a small number of the strongest dependencies explicitly (e.g., billing dispute → support sentiment → churn) as a Bayesian network rather than assuming full independence captures this cascade and typically improves predictive accuracy, at the cost of needing more data to estimate the conditional probability tables reliably.

3.5 Handling Uncertainty in Customer Behaviour

- Probabilistic (not binary) output: the system reports a churn probability rather than a hard yes/no label, letting the business calibrate intervention aggressiveness to risk tolerance and cost of retention offers.
- Bayesian updating absorbs new evidence gracefully — a single unusual month of low usage does not overwhelm a long history of stable behaviour, because the posterior blends prior belief with new likelihood rather than resetting.
- Confidence/credible intervals around the churn estimate can flag customers whose behaviour is genuinely ambiguous (wide interval) versus confidently at risk (narrow, high interval), directing human review to the ambiguous cases.
- Missing or noisy feedback data (e.g., customers who never respond to satisfaction surveys) is handled naturally in the Bayesian framework by simply omitting that likelihood term rather than requiring imputation.

3.6 Effectiveness for Retention Strategy

A well-calibrated probabilistic churn model lets the company rank customers by expected loss (churn probability × customer lifetime value) and allocate retention budget accordingly — targeting high-risk, high-value customers with proactive offers while leaving low-risk customers untouched. The main limitations are that Naïve Bayes' independence assumption can under- or over-estimate risk when feature dependencies are strong, that priors must be kept current as the overall churn rate drifts (concept drift), and that probabilistic churn scores are only as actionable as the retention interventions available — accurate prediction alone does not retain customers without an effective, timely response.

4. **A smart healthcare system is designed to monitor patients in real-time using wearable sensors. Using Bayesian Belief Networks and probability learning, develop a model to detect anomalies in vital signs. Analyze how conditional dependencies between physiological parameters are modeled and how uncertainty due to sensor noise is handled. Evaluate system performance and discuss challenges in real-time deployment.**
   Answer :

Real-Time Patient Anomaly Detection Using Bayesian Belief Networks

4.1 Problem Framing

A smart healthcare system continuously ingests vital-sign streams from wearable sensors — heart rate, blood oxygen (SpO2), blood pressure, respiration rate, body temperature — and must flag anomalous patterns (e.g., early signs of sepsis, arrhythmia, or respiratory distress) in real time, while distinguishing genuine physiological events from sensor noise or motion artefacts.

4.2 System Design: Bayesian Belief Network (BBN)

Why a BBN rather than Naïve Bayes: Vital signs are physiologically interdependent — for instance, heart rate and blood pressure are coupled through cardiac output, and respiration rate influences blood oxygen levels. A Bayesian Belief Network explicitly represents these conditional dependencies as a directed acyclic graph, with each node's conditional probability table (CPT) specifying P(node | parents), which is far more faithful to physiology than the flat independence assumption used in the spam and churn systems above.

Network structure: Nodes represent both observed sensor readings and latent health-state variables (e.g., "cardiovascular stress", "respiratory distress"). Edges encode causal or statistical dependencies established from clinical domain knowledge and/or learned from historical patient data (structure learning), while parameters (the CPTs) are learned from labelled or semi-labelled historical episodes using probability learning techniques such as maximum likelihood or Bayesian parameter estimation with informative clinical priors.

Inference: Given a new set of sensor readings, the network performs probabilistic inference (e.g., belief propagation, variational inference, or particle-filter methods for streaming data) to compute the posterior probability of each latent health state, and an alarm is raised when this posterior crosses a clinically calibrated threshold.

4.3 Modelling Conditional Dependencies

- Direct physiological links are encoded as edges — e.g., Respiration Rate → SpO2, Heart Rate ↔ Blood Pressure (via a shared "cardiac output" latent node), Temperature → Heart Rate (fever-driven tachycardia).
- Conditioning on the latent health-state node lets the network explain a cluster of correlated abnormal readings with a single underlying cause, rather than treating each abnormal vital sign as an independent alarm — reducing false positives caused by, for example, a benign temporary heart-rate spike during movement.

- The network can be queried in both directions: forward (predict expected vital signs given a suspected condition) and diagnostic/backward (infer the most probable underlying condition given the observed vitals), which is a key advantage of a full generative Bayesian network over a purely discriminative classifier.

4.4 Handling Sensor Noise and Uncertainty

Noise modelling: Each sensor reading is treated as a noisy observation of an underlying true physiological value, typically modelled with an explicit observation-noise distribution (e.g., a Gaussian around the true value); this is standard in Bayesian filtering approaches such as Kalman filters or particle filters used for real-time vital-sign smoothing before the BBN inference step.

Sensor fusion: Where multiple sensors provide correlated information (e.g., PPG-derived heart rate and ECG-derived heart rate), Bayesian fusion combines them weighted by their respective reliabilities, rather than trusting either sensor blindly.

Motion/artefact detection: An auxiliary node for "signal quality" or "motion artefact" can be added to the network so that low-confidence readings are automatically down-weighted in the anomaly inference rather than treated as equally reliable evidence.

Calibrated alerting: Because the network outputs a probability rather than a hard flag, the alerting threshold can be tuned per patient or per condition to balance sensitivity (catching real deterioration early) against alarm fatigue (excessive false alerts to clinical staff), which is a well-documented risk in continuous monitoring systems.

4.5 Performance Evaluation and Real-Time Deployment Challenges

Dimension	Consideration
Detection accuracy	Evaluated via sensitivity/specificity and precision-recall curves against clinician-labelled deterioration events; false negatives (missed deterioration) are far costlier than false positives, so thresholds are typically biased toward sensitivity
Latency	Inference must complete within a small time budget (seconds) for genuine real-time monitoring; exact BBN inference can be intractable for large networks, so approximate inference (loopy belief propagation, sampling) is often required
Alarm fatigue	Over-sensitive thresholds erode clinician trust; the network's probabilistic output allows graded alerts (informational vs urgent) instead of binary alarms
Data drift / individual variability	Baseline vitals vary by patient (age, fitness, comorbidities); the system benefits from per-patient Bayesian updating so priors adapt to each patient's personal baseline rather than a fixed population prior
Missing/intermittent data	Wearables disconnect or lose signal; BBNs naturally handle missing nodes by marginalising over them, unlike models requiring a complete fixed feature vector
Interpretability	Clinicians need to understand why an alert fired; the explicit

	graph structure of a BBN supports explanation ("elevated heart rate combined with falling SpO2 suggests respiratory distress") in a way opaque black-box models do not

Overall, a Bayesian Belief Network is well matched to real-time vital-sign monitoring because it (1) encodes genuine physiological dependencies rather than assuming independence, (2) natively represents and propagates uncertainty from noisy sensors, and (3) produces calibrated, explainable probabilities suitable for clinically graded alerting. The principal deployment challenges are computational latency for exact inference at scale, adapting priors to individual patients, and tuning the sensitivity/alarm-fatigue trade-off in collaboration with clinical staff.

RUBRICS FOR CALCULATION

Criteria	Weightage	Level 4 – Exemplary (5 Marks)	Level 3 – Proficient (4 Marks)	Level 2 – Developing (2–3 Marks)	Level 1 – Beginning (0–1 Mark)
Conceptual Understanding	**30**	Demonstrates clear and complete understanding of the concept	Good understanding with minor gaps	Basic understanding with some misconceptions	Poor or incorrect understanding
Accuracy of Answer	**20**	Completely correct answer with no errors	Mostly correct with minor errors	Partially correct with noticeable errors	Incorrect or irrelevant answer
Application of Knowledge	**20**	Correctly applies concepts to solve the question/problem	Applies concepts with minor mistakes	Limited or partially correct application	No or incorrect application
Completeness of Response	**15**	Covers all required parts of the question	Covers most parts with minor omissions	Covers only some parts	Incomplete or missing response
Clarity & Presentation	**15**	Clear, concise, and well-organized answer	Understandable with minor issues	Somewhat unclear or poorly structured	Difficult to understand